Related Work

Methodology

The literature review was conducted following a problem-driven approach. Starting from the practical problem of routing vehicles for street cleaning services, we identified that the underlying structure corresponds to an Arc Routing Problem (ARP). To understand the landscape of ARP variants, we first consulted key surveys and reviewed papers on arc routing [2]. From these reviews, we traced forward by examining the papers that cited them, as well as the references within them. We then narrowed our selection to studies whose problem variants share structural characteristics with ours — specifically, variants of the Rural Postman Problem (RPP) and its extensions involving mixed graphs, profit collection, time constraints, multiple vehicles, and heterogeneous fleets. This allowed us to systematically position our problem within the existing literature and identify the combination of features that distinguishes it from prior work.

Literature review

Literature on Arc Routing Problems (ARP) has been steadily growing, as many real-world applications can be formulated as an ARP [2]. These applications include street cleaning, salt spreading, snow clearing, street mapping, road maintenance, plotters, printers, cutting machines, newspaper delivery, postal services, and waste collection.

At its core, our problem is an ARP: the city is represented as a graph where streets correspond to edges/arcs and junctions correspond to nodes, and vehicles must perform services along a subset of these edges.

More specifically, our problem is a variant of the Rural Postman Problem (RPP) [3], one of the most well-studied special cases of the ARP. Unlike the Chinese Postman Problem (CPP) [19], which requires traversing all edges in a graph, the RPP requires traversing only a subset of them. Similarly, our problem distinguishes between required edges that must be serviced, non-required edges that serve only as connectors, and optional profitable edges that yield a score when visited.

Since our graph contains both directed edges (one-way streets) and undirected edges (two-way streets), the problem falls under the Mixed Rural Postman Problem (MRPP) [4], a special case of the RPP defined on mixed graphs. However, our additional constraints — including vehicle heterogeneity, edge diversity, and time limitations — distinguish it from the standard MRPP.

The presence of optional profitable edges relates our problem to the Prize-collecting Rural Postman Problem (PRPP) [5], in which there are no required edges and instead a profit is collected for each edge visited for the first time. Our variant differs in that it combines both required edges and profitable edges: in addition to the streets that must be serviced, there exists a

set of optional edges from which a subset must be selected to maximize the total score collected. This aspect aligns with the broader class of Arc Routing Problems with Profits.

Furthermore, our problem incorporates a hard time constraint, as each vehicle must complete its route and return to the depot within an eight-hour window. Time-constrained arc routing problems have been studied in the context of winter gritting operations [7], among others.

Finally, our problem involves multiple vehicles with different capabilities. The use of more than one vehicle corresponds to the k-vehicle Arc Routing Problem (k-ARP) [1], [7], [8], [12], while the heterogeneity of the fleet — where vehicles differ in their operational characteristics — aligns with the Heterogeneous Fleet Arc Routing Problem (HARC) [14].

We have defined seven major features that define the structure of the ARP. Using these features we compare different variants of ARP to show that the new combinations of these features gives a new variant of the ARP for street cleaning.

Features

1. Required Visit
   The problem requires that only a subset of edges/arcs of the graph are visited.
2. Mixed Graph
   The graph consists of both undirected edges and directed arcs.
3. Time Constraint
   There is a strict limit on how long a vehicle can be out on the road. This includes both the time spent working (servicing a task) and the time spent driving between tasks.
4. Multiple Vehicle
   You are managing a fleet of vehicles, not just a single driver.
5. Depot Return
   Vehicles must start and end at the same location.
6. Profit Collection
   Not all streets are equal; some have "prizes" or value attached to them. The algorithm focuses on collecting these values in addition that it has to visit the required edges.
7. Heterogeneous Fleet
   The vehicles are not identical, they can differ in terms of capacity, speed and other characteristics.

Study	Required Visit	Mixed Graph	Time Constraint	Multiple Vehicle	Depot Return	Profit Collection	Heter. Fleet	Total
[21] Huang et al.	+	-	-	-	-	-	-	1
[11] Carola et al.	+	+	-	-	-	-	-	2
[20] Kraiem et al.	+	-	+	-	-	-	-	2
[8] Nielsen	+	-	-	+	-	-	+	3
[9] Eglese & Murdock	+	-	+	-	+	-	-	3
[22] Kraiem et al.	+	+	+	+	-	-	-	4
[10] Bodin & Kursh	+	-	+	+	+	-	-	4
[7] Eglese	+	-	+	+	+	-	+	5
Our Problem	+	+	+	+	+	+	+	7

Table 1. Comparison with Street Cleaning Routing Studies

Table 1. shows the gap between our variant of ARP and other variants of ARP that are in the domain of street cleaning. No other ARP variant on the street cleaning domain studies a case when all seven features are combined into one problem. Notably none of these problems have a Profit Collection feature. Most of them have a homogeneous fleet and their graph is not mixed.

Study	Required Visit	Mixed Graph	Time Constraint	Multiple Vehicle	Depot Return	Profit Collection	Heter. Fleet	Total
[3][6] RPP	+	-	-	-	+	-	-	2
[5] PCRPP	-	-	-	-	+	+	-	2
[18] CARPIF	+	-	-	-	+	-	-	2
[4] MRPP	+	+	-	-	+	-	-	3
[17] CARP	+	-	-	+	+	-	-	3
[1] SVRP	-	+	+	+	-	+	-	4
[13] TOARP	+	-	+	+	+	+	-	5
[14] CLARPD	+	-	+	+	+	-	+	5
[16] FCARP	+	-	+	+	+	-	+	5
[15] CGRP	+	+	+	+	+	-	+	6
[12] K-OMP	+	+	+	+	+	+	-	6
Our Problem	+	+	+	+	+	+	+	7

Table 2. Comparison with (Non-Domain Specific Street Cleaning) Arc Routing Problem Variants

With Table 2 we want to show the gap between our variant of ARP and other variants of ARP that are close in features. We include the SVR problem even though it is not an ARP because of its closeness to our variant. We also include some major variants of the related literature.

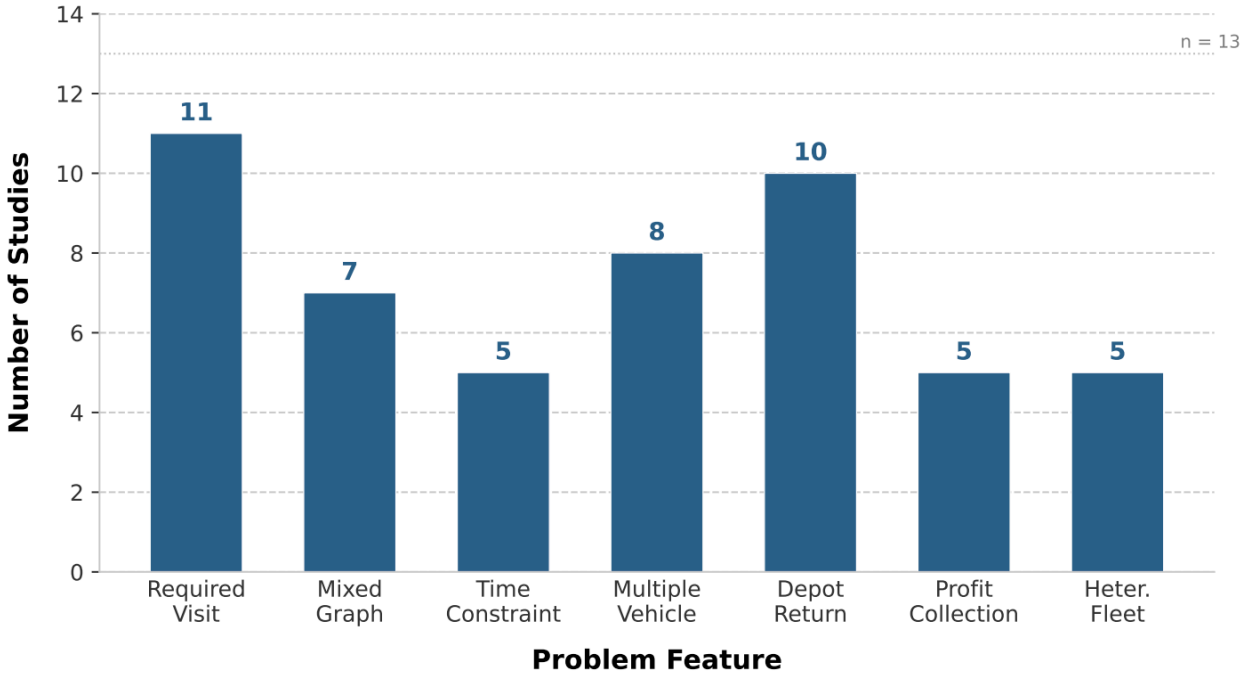


Figure 1. Feature frequency distribution

What we understand from Figure 1 is that the features are used in the previous problems, they are not unique to our study. The most studied and used features are the required visit of edges and the return to the depot of the vehicles. While the least studied and used features are the time constraints, profit collection and the heterogenous fleet. Our study uses a combination of all these features making it unique accoros the different ARP variants.

References

[1] Google Hash Code Archive. (2014). Problem statement for Main Round Street View Routing (Hash Code 2014 Final Round). GitHub. https://github.com/pierreavn/google-hashcode-archive/blob/main/archive/2014/final/hashcode_2014_final_round.pdf

[2] Corberán, Á.; Eglese, R.; Hasle, G.; Plana, I.; Sanchís Llopis, JM. (2021). Arc routing problems: A review of the past, present, and future. Networks. 77(1):88-115. https://doi.org/10.1002/net.21965

[3] Orloff, C. S. (1974). A fundamental problem in vehicle routing. *Networks*, *4*(1), 35-64.

[4] Corberán, A., Martí, R., & Romero, A. (2000). Heuristics for the mixed rural postman problem. *Computers & operations research*, *27*(2), 183-203.

[5] Aráoz, Fernández & Meza (2009) - "Solving the Prize-collecting Rural Postman Problem"

[6] Pearn, W. L., & Wu, T. C. (1995). Algorithms for the rural postman problem. *Computers & Operations Research*, *22*(8), 819-828.

[7] Eglese, R. W. (1994). Routeing winter gritting vehicles. *Discrete applied mathematics*, *48*(3), 231-244.

[8] Nielsen, L. D. (2019). Winter Gritting Routes with Multiple Visits.

[9] Eglese, Richard W.; Murdock, Heather . (1991). *Routeing Road Sweepers in a Rural Area. Journal of the Operational Research Society, 42(4), 281–288.* doi:10.1057/jors.1991.66

[10] Bodin, L.D., Kursh, S.J., 1978.
A Computer-Assisted System for the Routing and Scheduling of Street Sweepers.
Operations Research 26, 525–537.

[11] Carola A. Blazquez, Alejandra Beghelli & Veronica P. Meneses
(2012) A novel methodology for determining low-cost fine particulate matter street sweeping routes, Journal of the Air & Waste Management Association, 62:2, 242-251, DOI: 10.1080/10473289.2011.646048

[12] PMCARP — Benavent, Corberán, Gouveia, Mourão & Pinto (2015) "Profitable mixed capacitated arc routing and related problems", TOP 23, 244–274

[13] TOARP — Archetti, Corberán, Plana, Sanchis & Speranza (2014) "The Team Orienteering Arc Routing Problem", Transportation Science 48(3), 442–457

[14] CLARPD — Mirzaei-Khafri, Bashiri, Soltani & Khalilzadeh (2019) "A mathematical model for the capacitated location-arc routing problem with deadlines and heterogeneous fleet", Transport 34(6), 692–707

[15] CGRP — Pandit & Muralidharan (1995) "Capacitated General Routing Problem"

[16] Ulusoy, G. (1985). The fleet size and mix problem for capacitated arc routing. *European Journal of Operational Research*, *22*(3), 329-337.

[17] Golden & Wong (1981) - "Capacitated Arc Routing Problems (CARP)"

[18] Ghiani, G., Improta, G., & Laporte, G. (2001). The capacitated arc routing problem with intermediate facilities. *Networks: An International Journal*, *37*(3), 134-143.

[19] Mei-Ko, K. (1962). Graphic programming using odd or even points. *Chinese Math*, *1*, 237-277.

[20] Kraiem, A., Audy, J.F., Lamghari, A., 2025.
Mixed integer linear programming model for a multi-depot arc routing problem with different arc types and flexible assignment of end depot.
Transportation Research Procedia 82, 1109–1119.

[21] Huang, Y., Mou, W., Lan, J., Luo, F., Wu, K., Lu, S., 2024.
Multi-Objective Path Planning for Unmanned Sweepers Considering Traffic Signals: A Reinforcement Learning-Enhanced NSGA-II Approach.

[22] Kraiem, A., Audy, J.F., Lamghari, A., 2024.
Adaptive large neighbourhood search for the multi-depot arc routing problem with flexible assignment of end depot and different arc types.

Abbreviations

RPP — Rural Postman Problem

PCRPP — Prize-Collecting Rural Postman Problem

SVRP — Street View Routing Problem

MRPP — Mixed Rural Postman Problem

TOARP — Team Orienteering Arc Routing Problem

CLARPD — Capacitated Location–Arc Routing Problem with Deadlines and Heterogeneous Fleet

CGRP — Capacitated General Routing Problem

FCARP — Fleet Size and Mix Problem for Capacitated Arc Routing

CARP — Capacitated Arc Routing Problem

CARPIF — Capacitated Arc Routing Problem with Intermediate Facilities

K-OMP — The team orienteering mixed capacitated arc routing problem